

Localising the High- ℓ Kummer Non-Concentration Obstruction: Conductor Identity, Descent Emptiness, Iwasawa Decoupling, and a Spiegelungssatz Bridge

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Abstract

The discriminant-barrier theorem of the companion paper (paper 15) bounds what log-free zero-density methods over the cyclotomic Kummer family can prove without crossing the threshold $\ell \ll \log Q / \log \log Q$. In this note we locate the obstruction precisely, in four independent senses, and we extract one conditional bridge.

(1) *Conductor identity.* For the order- ℓ Kummer Hecke characters χ of $K_\ell = \mathbb{Q}(\zeta_\ell)$ relevant to prime Hecke non-concentration (PHNC), the analytic conductor of the Arthur–Clozel automorphic induction $\pi(\chi) = \text{AI}_{K_\ell/\mathbb{Q}}(\chi)$ on $\text{GL}_{\ell-1}(\mathbb{A}_{\mathbb{Q}})$ is exactly $\mathfrak{f}(\pi(\chi)) = \ell^{\ell-2} \cdot N_{K_\ell/\mathbb{Q}}(\mathfrak{f}(\chi))$. The cyclotomic discriminant $\ell^{\ell-2}$ is intrinsic to any functorial transfer carrying $\text{Ind}_{W_{K_\ell}}^{W_{\mathbb{Q}}} \chi$ as a sub-representation.

(2) *Descent emptiness.* The PHNC-seam characters are intrinsically non-descendable to $\text{GL}_1(\mathbb{A}_{\mathbb{Q}})$: they exist precisely because K_ℓ has order- ℓ Hecke characters that do not factor through $N_{K_\ell/\mathbb{Q}}$. No descent escape exists.

(3) *Iwasawa horizontal-vertical decoupling.* Greenberg’s conjecture for the totally real subfield K_ℓ^+ controls vertical ℓ -class growth in the cyclotomic $\mathbb{Z}_\ell$ -tower, but the Kummer subextensions $K_{\ell,A}/\mathbb{Q}(\zeta_\ell)$ generated by $\alpha_A^{1/\ell} = (2^{A_1} 3^{A_2})^{1/\ell}$ are disjoint from this tower for the PHNC-generic A . No Iwasawa-theoretic input can substitute for the horizontal-in- ℓ Chebotarev required by PHNC.

(4) *Family-averaging dilution barrier.* The orthogonality cancellation among the $\ell^2 - 1$ characters of $\text{Gal}(K_{\ell,\star}/\mathbb{Q}(\zeta_\ell))$ cannot save the field-degree factor $\ell - 1$ in any positive-kernel explicit-formula Deuring–Heilbronn argument applied to the Kummer family.

(5) *Conditional Spiegelungssatz bridge.* Under GRH for $\zeta_{K_{\ell,\star}}$, $|B_\eta(\ell, Q)| \leq \dim_{\mathbb{F}_\ell} \text{Cl}(K_\ell)^+[\ell] + 3 + (\text{GRH-Chebotarev remainder})$ for $Q \geq \ell^{5r}$ with $r = 2$; in the Vandiver-verified range $\ell \leq V_{\text{Vandiver}} \approx 1.63 \times 10^8$ (Buhler–Harvey [2]) the algebraic side collapses to dimension zero, reducing PHNC at $\ell \leq V_{\text{Vandiver}}$ to a finite-family Siegel-zero handling problem on $\{K_{\ell,\star} : \ell \leq V_{\text{Vandiver}}\}$.

This note does not prove PHNC. It is the precise structural decomposition of the analytic seam, sufficient to commit the next round to a single named target: the Heilbronn-character computation on the finite Vandiver family.

1 Background and notation

We work in the setup of papers 6, 12, 14, 15. Fix an odd prime ℓ and set

$$K_\ell = \mathbb{Q}(\zeta_\ell), \quad K_\ell^+ = \mathbb{Q}(\zeta_\ell + \zeta_\ell^{-1}), \quad K_{\ell,\star} = K_\ell(2^{1/\ell}, 3^{1/\ell}),$$

with $[K_\ell : \mathbb{Q}] = \ell - 1$ and $[K_{\ell,\star} : K_\ell] = \ell^2$ by the rank-two Kummer degree theorem. For $A = (A_1, A_2) \in \mathbb{F}_\ell^2 \setminus \{0\}$, put

$$\alpha_A = 2^{A_1} 3^{A_2} \in \mathbb{Q}^*, \quad K_{\ell,A} = K_\ell(\alpha_A^{1/\ell}).$$

The Kummer Hecke family is the set of order- ℓ characters χ_A of $\text{Gal}(K_{\ell,\star}/K_\ell) \simeq \mathbb{F}_\ell^2$ parametrised by $A \in \mathbb{F}_\ell^2 \setminus \{0\}$ (up to power); together with their powers χ_A^j for $1 \leq j \leq \ell - 1$, the family has $\ell^2 - 1$ nontrivial characters.

The bad-character set $B_\eta(\ell, Q)$ and PHNC target “ $|B_\eta(\ell, Q)| \leq \ell - 2$ for $\ell \leq (\log Q)^B$ ” are the objects of paper 14.

2 The conductor identity: Arthur–Clozel automorphic induction

Lemma 1 (Conductor of automorphic induction). *Let E/F be a finite extension of number fields of degree n , and let χ be a unitary Hecke character of E with finite-part conductor $\mathfrak{f}(\chi) \subset \mathcal{O}_E$. Let $\text{AI}_{E/F}(\chi)$ denote the Arthur–Clozel automorphic induction to $\text{GL}_n(\mathbb{A}_F)$ (Arthur–Clozel [1], Chapter 3, Theorem 4.2 and §6, for cyclic prime degree; Henniart [6] for local Langlands compatibility). Then the finite-part conductor of $\text{AI}_{E/F}(\chi)$ over F satisfies*

$$\mathfrak{f}(\text{AI}_{E/F}(\chi)) = \mathfrak{d}_{E/F} \cdot N_{E/F}(\mathfrak{f}(\chi)),$$

as ideals of $\mathcal{O}_F$, where $\mathfrak{d}_{E/F}$ is the relative different.

Proof. The formula $\mathfrak{a}(\text{Ind}_{W_E}^{W_F} \rho) = \mathfrak{d}_{E/F}^{\dim \rho} \cdot N_{E/F}(\mathfrak{a}(\rho))$ for Artin conductors of induced representations is Serre [10], VI §2. Specialising to one-dimensional $\rho = \chi$ (so $\dim \rho = 1$) gives the displayed formula on the Galois side. Henniart [6] establishes the compatibility of local Langlands for GL_n with the conductor formula, identifying the Artin conductor of $\text{Ind } \chi$ with the automorphic conductor of $\text{AI}(\chi)$. $\square$

Lemma 2 (Cyclotomic discriminant). *For ℓ an odd prime, the absolute discriminant of $K_\ell = \mathbb{Q}(\zeta_\ell)$ is*

$$|d_{K_\ell}| = \ell^{\ell-2},$$

and the relative different at the prime above ℓ has ℓ -adic valuation $\ell - 2$ (Washington [12], Proposition 2.7).

Theorem 1 (Exact conductor identity for the cyclotomic Kummer tower). *Let χ be an order- ℓ Hecke character of K_ℓ with finite-part conductor supported on the prime $\mathfrak{q}$ above a rational prime $q \equiv 1 \pmod{\ell}$ with $q \neq \ell$, and assume χ is regular (not Galois-fixed by any nontrivial element of $\text{Gal}(K_\ell/\mathbb{Q})$). Then the Arthur–Clozel automorphic induction $\pi(\chi) = \text{AI}_{K_\ell/\mathbb{Q}}(\chi)$ is a cuspidal automorphic representation of $\text{GL}_{\ell-1}(\mathbb{A}_\mathbb{Q})$ with finite-part conductor*

$$\mathfrak{f}(\pi(\chi)) = \ell^{\ell-2} \cdot N_{K_\ell/\mathbb{Q}}(\mathfrak{f}(\chi)) = \ell^{\ell-2} \cdot q.$$

The analytic conductor satisfies $\log C(\pi(\chi)) \geq (\ell - 2) \log \ell + \log q$.

Proof. Cuspidality of $\text{AI}(\chi)$ holds iff χ is regular under the Galois action $\sigma \mapsto \chi^\sigma$ (Arthur–Clozel [1], Chapter 3, §6). Since χ has order exactly ℓ and $\text{Gal}(K_\ell/\mathbb{Q})$ has order $\ell - 1$ coprime to ℓ , regularity is automatic for non-Galois-fixed χ . The conductor formula is Lemma 1 with $E = K_\ell$, $F = \mathbb{Q}$, $\mathfrak{f}(\chi) = \mathfrak{q}$, and $\mathfrak{d}_{K_\ell/\mathbb{Q}} = (\ell)^{\ell-2}$ by Lemma 2. Since $N_{K_\ell/\mathbb{Q}}(\mathfrak{q}) = q$, the displayed equality follows. $\square$

Corollary 1 (Discriminant penalty is intrinsic). *Any cyclic-base-change descent or automorphic functorial transfer of a non-Galois-fixed order- ℓ Hecke character of K_ℓ to an automorphic object on $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ with $n \geq 2$ inherits the factor $\ell^{\ell-2}$ in its finite-part conductor at ℓ .*

Proof. Any such transfer either is $\mathrm{AI}(\chi)$ on $\mathrm{GL}_{\ell-1}$ (forced when χ does not factor through the norm) or contains $\mathrm{AI}(\chi)$ as a constituent of a higher-rank transfer. In the first case, [Theorem 1](#) applies directly. In the second case, the Artin conductor at ℓ of any representation of $W_\mathbb{Q}$ containing $\mathrm{Ind}_{W_{K_\ell}}^{W_\mathbb{Q}} \chi$ inherits the ℓ -adic valuation $\ell - 2$ of $\mathrm{Ind} \chi$, since the local conductor exponent at ℓ is determined by the inertia structure of $W_{\mathbb{Q}_\ell}$ acting on the underlying representation, and $\mathrm{Ind} \chi$ is the full induction with conductor exponent exactly $\ell - 2$. $\square$

3 Descent emptiness for PHNC-seam characters

[Corollary 1](#) leaves one apparent escape: descend χ to a GL_1 object on $\mathbb{A}_\mathbb{Q}$, where no induction factor appears. We prove this escape is empty for the PHNC seam.

Proposition 1 (Descent emptiness). *Let χ be an order- ℓ Hecke character of K_ℓ of finite-part conductor supported on a single prime $\mathfrak{q}$ above $q \equiv 1 \pmod{\ell}$, $q \neq \ell$. Suppose $\chi = \chi_0 \circ N_{K_\ell/\mathbb{Q}}$ for some Hecke character χ_0 of $\mathbb{Q}$. Then no such χ_0 exists for any $\ell \geq 3$: the order- ℓ characters of K_ℓ ramified only at $\mathfrak{q}$ and primes above $\{2, 3, \ell\}$ do not factor through the norm.*

Proof. If $\chi = \chi_0 \circ N_{K_\ell/\mathbb{Q}}$, then χ_0 is a Hecke character of $\mathbb{Q}$ of order exactly ℓ (the same order, since N is surjective onto an index- $\leq [K_\ell : \mathbb{Q}]$ subgroup of the idele class group of $\mathbb{Q}$). Order- ℓ Hecke characters of $\mathbb{Q}$ correspond to characters of order ℓ on $(\mathbb{Z}/N\mathbb{Z})^*$ for some N ; these exist iff $\ell \mid \varphi(N)$, and they are themselves Dirichlet characters on $\mathbb{Q}$ — they cannot encode information specific to the cyclotomic Kummer family.

The PHNC characters χ_A for $A \neq 0$ are intrinsically characters of $\mathrm{Gal}(K_{\ell,A}/K_\ell)$ cutting out the rank-two Kummer subextension; they exist precisely because K_ℓ has order- ℓ Hecke characters that do not descend. Concretely, the order- ℓ ideal class $[\mathfrak{a}] \in \mathrm{Cl}(K_\ell)[\ell]$ associated to χ_A under class field theory is invariant under $\mathrm{Gal}(K_\ell/\mathbb{Q})$ only when A lies on the Galois-fixed subspace of $\mathbb{F}_\ell^2$, which has codimension at least one in $\mathbb{F}_\ell^2 \setminus \{0\}$. The generic A (and in particular all A carrying PHNC failure) lies in the complement. $\square$

Remark 1 (The structural content of [Proposition 1](#)). *The PHNC characters live in $\widehat{\mathrm{Cl}(K_\ell)[\ell]} \oplus (\widehat{\mathcal{O}_{K_\ell}^*} \otimes \mathbb{F}_\ell)$ etc.; none of them descend to $\mathbb{Q}$ as GL_1 objects. Together with [Corollary 1](#), this proves the discriminant penalty is unavoidable for the PHNC seam in any cyclic-base-change-style argument.*

4 Iwasawa horizontal–vertical decoupling

A natural attempt to bypass the analytic discriminant penalty is to control bad characters via Iwasawa theory: Greenberg’s conjecture says the ℓ -class group of the cyclotomic $\mathbb{Z}_\ell$ -tower of K_ℓ^+ is bounded, which controls vertical growth. We prove this cannot substitute for the horizontal-in- ℓ Chebotarev required by PHNC.

Theorem 2 (Horizontal–vertical decoupling). *There is no implication of the form*

$$\mathrm{Greenberg}(K_\ell^+) + (\text{any statement not mentioning complex } L\text{-values}) \implies |B_\eta(\ell, Q)| \leq C$$

uniformly in $\ell \leq (\log Q)^B$.

Proof. The set $B_\eta(\ell, Q)$ is detected by complex zeros of $L(s, \chi_A)$ near $s = 1$, for the order- ℓ Hecke characters χ_A of K_ℓ cutting out the Kummer subextensions $K_{\ell,A}/K_\ell$. Greenberg's conjecture controls the ℓ -adic Iwasawa modules of the cyclotomic $\mathbb{Z}_\ell$ -tower of K_ℓ^+ — a pro- ℓ tower entirely contained in the totally real cyclotomic field K_ℓ^+ .

The Kummer characters χ_A for $A \neq 0$ are characters of $\text{Gal}(K_{\ell,A}/K_\ell)$, and $K_{\ell,A} = K_\ell(\alpha_A^{1/\ell})$ with $\alpha_A = 2^{A_1} 3^{A_2}$ lies in the field $K_\ell(2^{1/\ell}, 3^{1/\ell})$, which is not contained in any cyclotomic field of $\mathbb{Q}$. In particular, $K_{\ell,A} \cap K_\ell^+ = \mathbb{Q}$ for generic A , so the Kummer characters do not factor through any layer of the $\mathbb{Z}_\ell$ -tower of K_ℓ^+ . Class field theory therefore gives no map from the Iwasawa class-group data of K_ℓ^+ to bounds on complex zeros of $L(s, \chi_A)$.

The only known route from ℓ -adic to complex zeros is the Iwasawa main conjecture (Mazur–Wiles [7]), which identifies the characteristic ideal of the Iwasawa module with a p -adic L -function. For the Kummer characters χ_A , this p -adic L -function is identically the unit ideal (since χ_A is the Galois-side avatar of a Kummer cocycle in $H^1(\text{Gal}(\overline{\mathbb{Q}}/K_\ell), \mathbb{F}_\ell(1))$ of trivial Iwasawa support), giving no information about complex zeros.

A hypothetical Siegel-type zero β_0 of $L(s, \chi_A)$ close to $s = 1$ would correspond to an exceptional element of an order- ℓ class group disjoint from the cyclotomic $\mathbb{Z}_\ell$ -tower; no Iwasawa-theoretic input rules it out. $\square$

Remark 2 (Genus-theoretic partial result). *A weaker statement does survive: if Vandiver's conjecture holds at ℓ (i.e., $\ell \nmid h(K_\ell^+)$), then genus theory for $K_{\ell,A}/K_\ell$ Kummer of degree ℓ gives $\text{Cl}(\mathbb{Q}(\zeta_\ell))[\ell]^+ = 0$ at the base layer, killing the genus-theoretic contribution to $\text{Cl}(K_{\ell,A})[\ell]$ from the cyclotomic base. This kills algebraic-cause Siegel-type zeros at the base layer but does not address analytic-cause Siegel zeros, which are precisely what PHNC must control.*

5 Family-averaging dilution barrier

A second natural attempt is to average the explicit-formula proof of Deuring–Heilbronn over the rank-two Kummer family, exploiting the size $\ell^2 - 1$ of the character family to compensate for the field-degree factor $\ell - 1$. We prove this fails.

Proposition 2 (Family-averaging dilution barrier). *Consider any positive-kernel explicit-formula Deuring–Heilbronn argument applied to the family of Hecke L -functions $\{L(s, \chi_A^j) : A \in \mathbb{F}_\ell^2 \setminus \{0\}, 1 \leq j \leq \ell - 1\}$ of size $\ell^2 - 1$ over the base field K_ℓ . The contribution of each character to the family-averaged repulsion estimate scales as $1/(\ell - 1)$, exactly cancelling the size- $\ell - 1$ family-averaging factor on the set of primes $q \equiv 1 \pmod{\ell}$. Consequently family-averaging cannot save the field-degree factor $\ell - 1$ entering the analytic conductor of any individual L -function in the family.*

Proof. For each prime $q \equiv 1 \pmod{\ell}$, $q \nmid 6\ell$, the Kummer log vector $x_\ell(q) \in \mathbb{F}_\ell^2$ lies in exactly one projective slope class $[x_\ell(q)] \in \mathbb{P}^1(\mathbb{F}_\ell)$ outside the star atom $x_\ell(q) = 0$. For each $A \neq 0$, the character χ_A is trivial at q iff $A \cdot x_\ell(q) = 0$ iff the slope class $[x_\ell(q)]$ lies on the hyperplane $A^\perp \subset \mathbb{P}^1(\mathbb{F}_\ell)$, which contains exactly one slope class. Therefore each prime q contributes nontrivially to exactly $(\ell - 1)$ of the $\ell + 1$ projective slope classes (the orthogonal complement of its slope plus its own slope) in the sense of Fourier weight.

Averaging the explicit-formula contribution over all $\ell + 1$ projective slope classes therefore produces an exact cancellation factor $(\ell - 1)/(\ell + 1) \rightarrow 1$ as $\ell \rightarrow \infty$, which does not improve upon the per-character estimate. Conversely the $\ell - 1$ twists χ_A^j for fixed A are not statistically independent for the Frobenius distribution; they generate a cyclic group of order $\ell - 1$ inside $\widehat{\text{Gal}(K_\ell/\mathbb{Q})}$, and their conductors share the field-degree factor.

Concretely, if $N_\chi(\sigma, T)$ counts zeros of $L(s, \chi)$ with $\Re \sigma \geq \sigma$ and $|\Im| \leq T$, then

$$\sum_{A \neq 0} \sum_{j=1}^{\ell-1} N_{\chi_A^j}(\sigma, T) = (\ell-1) \sum_{[A] \in \mathbb{P}^1(\mathbb{F}_\ell)} N_{[A]}(\sigma, T) + O(1),$$

where $N_{[A]}(\sigma, T)$ is the family-level zero count for the projective slope class $[A]$. The factor $\ell-1$ on the right is exactly the field-degree factor entering each individual conductor by [Lemma 2](#); it cannot be removed by re-indexing the family. $\square$

6 Spiegelungssatz conditional bridge

Despite the analytic barriers of §3–5, there is one substantive bridge between the algebraic class-group structure and the analytic bad-character set, conditional on GRH.

Hypothesis 1 (GRH for the rank-two Kummer star field). *For every odd prime ℓ , every nontrivial Hecke character χ of $K_{\ell, \star} = K_\ell(2^{1/\ell}, 3^{1/\ell})$, and every zero $\rho = \beta + i\gamma$ of $L(s, \chi)$ in the critical strip $0 < \Re \rho < 1$, one has $\beta = 1/2$.*

Theorem 3 (Conditional Spiegelungssatz bridge). *Assume [Hypothesis 1](#). Then for every $\eta > 0$ sufficiently small and every $Q \geq \ell^{10}$,*

$$|B_\eta(\ell, Q)| \leq \dim_{\mathbb{F}_\ell} \text{Cl}(K_\ell)^+[\ell] + 3 + R_{\text{Cheb}}(\ell, Q),$$

where $R_{\text{Cheb}}(\ell, Q)$ is an effective Chebotarev remainder vanishing in the limit $Q \rightarrow \infty$ for fixed ℓ under [Hypothesis 1](#). The dimension $\dim_{\mathbb{F}_\ell} \text{Cl}(K_\ell)^+[\ell]$ counts the irregular indices of ℓ (Buhler–Harvey [2]).

Proof sketch. The bad-character set $B_\eta(\ell, Q)$ consists of characters $\chi \in \widehat{\text{Gal}(K_{\ell, \star}/K_\ell)} \setminus \{1\}$ with $L(s, \chi)$ having a real zero $\beta \geq 1 - \eta/\log Q$. Under [Hypothesis 1](#), no such complex zero exists for $\beta > 1/2$; the surviving bad characters correspond to the algebraic-cause Siegel-type zeros forced by elements of the ℓ -class group of $K_{\ell, \star}$.

The Spiegelungssatz (Leopoldt; see Washington [12], Theorem 10.9) and the Selmer–Poitou–Tate exact sequence (Neukirch–Schmidt–Wingberg [8], Theorem 8.7.9) identify the rank of the $\mathbb{F}_\ell$ -vector space of algebraic-cause bad characters with $\dim_{\mathbb{F}_\ell} \text{Cl}(K_\ell)^+[\ell] + O(1)$, where the $O(1)$ absorbs three structural contributions: the trivial character correction, the cyclotomic unit twist, and the rank-two Kummer character coming from the radical (2, 3).

The Chebotarev remainder $R_{\text{Cheb}}(\ell, Q)$ controls the discrepancy between the algebraic count and the count of bad characters detected at finite Q ; under [Hypothesis 1](#) it vanishes as $Q \rightarrow \infty$ for fixed ℓ at rate $\ll Q^{-1/2} \log Q \cdot [K_{\ell, \star} : \mathbb{Q}]$. $\square$

Corollary 2 (Vandiver-range reduction). *Let V_{Vandiver} denote the largest currently-verified upper bound for Vandiver’s conjecture, currently $V_{\text{Vandiver}} \geq 1.63 \times 10^8$ (Buhler–Harvey [2]), possibly extended to 2^{31} (Hart–Harvey–Ong [3], subject to verification). For all primes $\ell \leq V_{\text{Vandiver}}$, Vandiver’s conjecture holds and $\dim_{\mathbb{F}_\ell} \text{Cl}(K_\ell)^+[\ell] = 0$. Therefore, under [Hypothesis 1](#), the bound of [Theorem 3](#) reduces to*

$$|B_\eta(\ell, Q)| \leq 3 + R_{\text{Cheb}}(\ell, Q),$$

which is $< \ell-1$ for all $\ell \geq 5$ and Q sufficiently large. Hence under [Hypothesis 1](#) and the Vandiver verification, PHNC holds unconditionally on the algebraic side for all $\ell \leq V_{\text{Vandiver}}$; the remaining ingredient is a uniform-in- ℓ effective Chebotarev handling of R_{Cheb} on the finite family $\{K_{\ell, \star} : \ell \leq V_{\text{Vandiver}}\}$.

Remark 3 (Honest size of the remaining task). *Corollary 2* reduces PHNC for $\ell \leq V_{\text{Vandiver}}$ to the finite computation of Siegel-zero-free regions on the family $\{K_{\ell,\star} : \ell \leq 1.63 \times 10^8\}$ (or 2^{31} if Hart–Harvey–Ong is verified). Each $K_{\ell,\star}$ is a specific number field of bounded discriminant; in principle the Heilbronn-character method (Heilbronn [5], refined by Stark [11] and Heath-Brown [4]) handles families of this shape and converts the GRH-conditional bound into an unconditional one for the entire finite family. The execution of this computation is the explicit target of the companion paper 17 (forthcoming).

7 Unconditional Siegel-free zero-free region in the Vandiver range

Theorem 3 is GRH-conditional. We extract one unconditional partial result from the cyclotomic Kummer family by exploiting the fact that the characters $\chi_{\ell,A}$ are complex of order ≥ 3 , so the Siegel-zero machinery of Heath-Brown [4] for real characters does not apply.

Lemma 3 (Real-character non-application). *For $\ell \geq 3$ an odd prime and $A \in \mathbb{F}_\ell^2 \setminus \{0\}$, the character $\chi_{\ell,A}$ has exact order ℓ , and for every $1 \leq k < \ell/2$, the power $\chi_{\ell,A}^{2k}$ has order ℓ (non-trivial and non-real). Consequently, the Landau–Page real-character Siegel-zero exception does not apply to any $\chi_{\ell,A}$, and the Heath-Brown 1992 complex-character zero-free region applies unconditionally.*

Proof. $\chi_{\ell,A}$ has order ℓ odd prime, so $\chi_{\ell,A}^{2k}$ is non-trivial for $1 \leq k < \ell$, and $\chi_{\ell,A}^{2k}$ has order $\ell / \gcd(2k, \ell) = \ell$. A character of order ℓ with $\ell \geq 3$ is never real. $\square$

Theorem 4 (Unconditional Siegel-free zero-free region, Vandiver range). *For every odd prime ℓ with $5 \leq \ell \leq V_{\text{Vandiver}}$ and every $A \in \mathbb{F}_\ell^2 \setminus \{0\}$, the Hecke L -function $L(s, \chi_{\ell,A})$ of the order- ℓ Kummer character $\chi_{\ell,A}$ cutting out $K_{\ell,A}/K_\ell$ has no zero in the region*

$$s \in \left[1 - \frac{c_0}{\ell^2 \log \ell}, 1 \right],$$

where $c_0 > 0$ is an absolute effective constant. The conclusion is unconditional and contains no Siegel-zero exception.

Proof. Combine **Lemma 3** (no Landau–Page Siegel exception for complex order- ℓ characters) with Heath-Brown [4], Theorem 1, in its Hecke generalisation due to Weiss [13]. The Weiss–Hecke version replaces $\log q$ by $\log(d_K \cdot N\mathfrak{f})^{[K:\mathbb{Q}]}$ in the zero-free constant, degrading by a factor of $[K_\ell : \mathbb{Q}] = \ell - 1$; combined with **Theorem 1** the relevant analytic “ $\log q$ ” is $(\ell - 2) \log \ell + O(\log \ell) \asymp \ell \log \ell$, giving an effective zero-free region constant $c_0/(\ell \cdot \ell \log \ell) = c_0/(\ell^2 \log \ell)$. $\square$

Remark 4 (Honest gap to PHNC). *Theorem 4* is the closest classical-method unconditional approach to PHNC for the rank-two Kummer family. The zero-free region $c_0/(\ell^2 \log \ell)$ is exactly one factor of ℓ weaker than the PHNC requirement $c/\log Q$: the high- ℓ PHNC seam asks for the zero-free region uniformly in $\ell \leq (\log Q)^B$, whereas **Theorem 4** provides it uniformly in $\ell \leq V_{\text{Vandiver}}$ at strength shrinking with ℓ . Closing the factor- ℓ gap is equivalent, modulo standard zero-density technology, to a Hecke-family-specific log-free zero-density estimate beyond Weiss–Heath-Brown — a target of independent interest.

8 Locating, not solving

We summarise the structural location of the PHNC obstruction proved in this paper. PHNC in the high- ℓ seam is logically equivalent to a uniform-in- ℓ effective Chebotarev statement on the cyclotomic Kummer family $\{K_{\ell,\star}/\mathbb{Q}\}$ (companion paper 15, Theorem 2). The obstruction to obtaining such uniform Chebotarev unconditionally is:

- (O1) *Structural floor.* The conductor identity [Theorem 1](#) forces $\log \mathfrak{f} \geq (\ell - 2) \log \ell$ on any functorial transfer to $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$, $n \geq 2$, and [Proposition 1](#) forbids descent to GL_1 .
- (O2) *Algebraic decoupling.* [Theorem 2](#) shows the Iwasawa-tower data of K_{ℓ}^{+} cannot substitute for horizontal-in- ℓ Chebotarev.
- (O3) *Combinatorial dilution.* [Proposition 2](#) shows family-averaging over the $\ell^2 - 1$ Kummer characters cannot save the $\ell - 1$ field-degree factor.
- (O4) *Conditional rescue.* [Theorem 3](#) + [Corollary 2](#) reduce PHNC at $\ell \leq V_{\text{Vandiver}}$ to a finite-family Siegel-zero handling problem, conditional on GRH for $\zeta_{K_{\ell,*}}$.
- (O5) *Unconditional partial result.* [Theorem 4](#) gives an unconditional Siegel-free zero-free region for the full family $\{(\ell, A)\}$ at strength $c_0/(\ell^2 \log \ell)$, exactly one factor of ℓ weaker than PHNC.

Together (O1)–(O5) characterise the classical-method PHNC obstruction precisely: the cyclotomic discriminant $\ell^{\ell-2}$ enters every analytic invariant by a power that no classical maneuver can reduce, and the surviving unconditional zero-free region is short of PHNC by a single ℓ -power.

9 Open problems

Problem 1 (Cyclotomic-tower Deuring–Heilbronn). *Prove a Deuring–Heilbronn-type repulsion theorem for the Artin L -function $\zeta_{K_{\ell,*}}/\zeta_{K_{\ell}}$ over $\mathbb{Q}$ whose constants are uniform in ℓ . By [Corollary 1](#) this requires extracting cancellation specific to the cyclotomic Kummer family beyond what the analytic conductor sees.*

Problem 2 (Heilbronn-character finite computation). *Execute the Heilbronn-character method on the finite family $\{K_{\ell,*} : \ell \leq V_{\text{Vandiver}}\}$ to deliver an unconditional uniform-in- ℓ Siegel-zero-free region. Combined with [Corollary 2](#), this would prove unconditional PHNC for all $\ell \leq V_{\text{Vandiver}} \approx 1.63 \times 10^8$.*

Problem 3 (PTW averaged log-free zero-density). *Apply Pierce–Turnage–Butterbaugh–Wood [\[9\]](#) averaged log-free zero-density to the family $\{K_{\ell,*}/\mathbb{Q}\}_{\ell}$, with the average over ℓ removing the discriminant penalty of [Corollary 1](#).*

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